

SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 1.1: Cell Structure — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 10	
Sub-Strand	Sub-Strand 1.1: Cell Structure
Driving Question	DRIVING QUESTION: "Why do cells in our body look so different from each other — and how does the shape of a cell determine the job it can do?"

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Healing Cut: Launching Our Investigation Into Why Cells Look Different	We observed a sequence of changes that happen when a finger is cut — bleeding slows, a clot forms, a scab develops, and new skin appears over several days	We learned that this familiar process is actually made up of many simultaneous events, and that scientists have found many different types of tiny structures working together during wound healing	We cannot yet explain why different structures look different or how each one knows what job to do — these are exactly the questions our Driving Question Board will help us answer over the next eleven lessons
Lesson 2 What Is a Cell? Introduction to Cell Theory	In this lesson, learners observed labelled diagrams of generalised animal cells, plant cells, and prokaryotic cells. They sorted structural feature cards into Prokaryote and Eukaryote categories and studied the ugali-kitchen analogy mapped onto cell structure components.	Learners learned the three tenets of cell theory (all living things are made of cells; the cell is the basic unit of life; all cells arise from pre-existing cells) and the contributions of Schleiden, Schwann, and Virchow. They learned the key structural differences between prokaryotic and eukaryotic cells, including the presence or absence of a membrane-bound nucleus and membrane-bound organelles.	Learners explained, using a Claim–Evidence–Reasoning paragraph, how Virchow's tenet ('all cells from pre-existing cells') accounts for the production of new skin cells during the healing of Amina's finger. They also explained their Version 1 cell model using a labelled diagram with a structural-function key and articulated the limitations of their current model.
Lesson 3 Prokaryotic vs. Eukaryotic Cells: Two Worlds Inside One Body	Learners observed two labelled diagrams — E. coli (prokaryote, linked to nyama choma food poisoning) and a human cheek cell (eukaryote) — and identified structural differences including the presence or absence of a membrane-bound nucleus, membrane-bound organelles, relative size, and cell wall composition.	Prokaryotic cells (such as E. coli bacteria) lack a membrane-bound nucleus and membrane-bound organelles; their DNA floats freely in a nucleoid region. Eukaryotic cells (such as human cheek cells) have a membrane-bound nucleus and membrane-bound organelles. Prokaryotes are generally much smaller (1–5 micrometres)	The bacteria that could infect Amina's cut (E. coli) are prokaryotes — structurally far simpler than her own body cells, which are eukaryotes. The presence of a membrane-bound nucleus in eukaryotic cells allows DNA to be protected and gene expression to be regulated in ways that support the complexity needed for wound healing. This

		than eukaryotes (10–100 micrometres). Both cell types contain ribosomes, but ribosomes are not membrane-bound in either.	structural difference explains why, even though bacteria and human cells are both alive, they behave so differently — and why Amina's white blood cells (eukaryotes) can carry out the complex task of recognising and destroying E. coli (prokaryote) cells in the wound.
Lesson 4 Plant Cell Ultrastructure — Every Organelle Has a Farming Job	A fully labelled plant cell diagram and (where available) electron micrograph images of a maize leaf chloroplast, a central vacuole in an onion cell, and a plant cell wall boundary.	Plant cells contain nine key organelles — cell wall, cell membrane, nucleus, chloroplasts, large central vacuole, mitochondria, endoplasmic reticulum, ribosomes, and Golgi apparatus — each with a specific structure-linked function, mapped to roles on a Rift Valley maize farm.	Unlike the animal cells (e.g., epidermal cells, platelets) involved in healing Kioni's cut, plant cells have organelles such as chloroplasts and a large central vacuole that reflect their unique job: making their own food and maintaining structural pressure. Structure and function are inseparable — a principle that applies equally to the specialised cells in the wound-healing phenomenon.
Lesson 5 Animal Cell Ultrastructure — Comparing Plant and Animal Cells	Learners received and analysed a labelled animal cell ultrastructure diagram alongside their Lesson 4 plant cell diagram. They completed a structured Comparison Table identifying organelles as present or absent in each cell type, and delivered group presentations of annotated diagrams with peer feedback.	Animal cells contain lysosomes (membrane-bound sacs of digestive enzymes) and centrioles (which organise cell division) — organelles absent in plant cells. Plant cells uniquely possess a cell wall, chloroplasts, and a large central vacuole. Nucleus, mitochondria, ribosomes, ER, Golgi apparatus, and cell membrane are shared. These structural differences reflect the different functional demands on each cell type.	Learners constructed Functional Justifications explaining why animal cells lack cell walls (flexibility needed for movement and shape change), chloroplasts (they obtain energy from food, not photosynthesis), and large vacuoles (no need for turgor pressure maintenance). They connected organelles to specific cells in the cut finger phenomenon — e.g., lysosomes in neutrophils digesting bacteria, centrioles enabling epidermal cell division during wound closure. Version 3 of the cumulative cell model was completed: a side-by-side colour-coded Plant vs. Animal diagram with a Model Revision Note.
Lesson 6 Microscopy Practical 1 — Observing Onion Epidermal Cells	Onion epidermal cells prepared as temporary wet mount slides stained with iodine: rectangular cells with visible cell wall, large central vacuole, nucleus stained dark brown/yellow by iodine, cytoplasm, and cell membrane pressed against the inner cell wall. NO chloroplasts observed.	Onion epidermal cells are plant cells that contain a cell wall, vacuole, and nucleus — but NO chloroplasts, because the onion bulb grows underground and is not exposed to light. This is evidence of cell specialisation: the onion epidermal cell is adapted for protection and storage, not photosynthesis. All plant cells share some features (cell wall, vacuole) but differ in which organelles they contain depending on their function.	The absence of chloroplasts in onion skin cells is not an error — it is evidence of specialisation. Cells only build and maintain the structures they need for their specific job. Just as the cells healing the Nairobi student's cut finger are each specialised (platelets for clotting, fibroblasts for collagen, epidermal cells for surface repair), the onion epidermal cell is specialised for its role. This confirms that having the same DNA does not mean every cell builds the same structures — gene expression is selective, and structure is determined by

			function.
Lesson 7 Microscopy Practical 2 — Human Cheek Cells: Observing Animal Cell Structure	Learners prepared temporary wet mounts of their own cheek (buccal) cells using methylene blue stain, focused them under a compound light microscope, and produced labelled biological drawings. They directly observed a cell membrane, cytoplasm, and nucleus in each cheek cell.	Animal cells (such as human cheek epithelial cells) contain a cell membrane, cytoplasm, and a membrane-bound nucleus, but lack a cell wall, chloroplasts, and a large central vacuole — structures present in the plant (onion) cells observed in Lesson 6. The absence of a rigid cell wall allows animal cells to adopt flexible, irregular shapes.	The structural differences between plant and animal cells are not random — they reflect the different needs and lifestyles of each cell type. The absence of a cell wall in animal cells allows them to change shape, which is directly relevant to the anchoring phenomenon: cells such as platelets and epidermal cells can flatten, extend, and migrate during wound healing precisely because they are not constrained by a rigid outer wall. This is one piece of the larger puzzle of how cell shape determines cell function.
Lesson 8 Cell Specialisation: Meaning, Significance, and Specialised Animal Cells	Learners examined annotated diagrams and micrographs of five specialised animal cells — red blood cells (biconcave, no nucleus), white blood cells (irregular shape, many lysosomes), nerve cells (long axon, myelin sheath), sperm cells (flagellum, mitochondria-packed midpiece), and muscle cells (many mitochondria, myofilaments) — and compared normal red blood cells with sickle cells using printed micrograph evidence.	Learners learned that cell specialisation is the process by which cells develop distinct structures that equip them for specific functions; that structural features such as the absence of a nucleus (RBC), a long axon (neuron), or a flagellum (sperm) are not random but are direct adaptations to function; and that when cell shape is disrupted — as in sickle-cell anaemia — function is catastrophically compromised, providing powerful evidence that shape determines function.	Learners explained how each of the five specialised animal cells contributes to the healing-cut phenomenon sequence — red blood cells deliver oxygen to repair tissue, white blood cells engulf bacteria at the wound site, nerve cells transmit pain signals, muscle cells enable movement that keeps the wound protected, and sperm cells (while not directly involved in healing) demonstrate the principle that extreme specialisation trades generalist survival for singular functional excellence. Learners used the Claim-Evidence-Reasoning framework to argue from the sickle-cell micrograph that cell shape is the physical basis of cell function.
Lesson 9 Specialised Plant Cells: Form Meets Function in the Bean Seedling	Learners observed detailed diagrams (and where available, microscope slides or projected images) of five specialised plant cells — root hair cell, guard cell, xylem vessel, phloem sieve tube, and palisade mesophyll cell — recording labelled scientific drawings and completing a three-column observation table of structural features, alongside a contextual comparison of stomatal aperture in dry Turkana vs. humid Kericho.	Learners learned that plant cells specialise just as animal cells do; that each of the five cells has a specific structural adaptation (e.g., the long extension of root hair cells for absorption, the hollow lignified lumen of xylem for water transport, the dense chloroplasts of palisade cells for photosynthesis) that directly enables its function; and that specialised plant cells work in coordinated systems — absorption → transport → photosynthesis → regulation — mirroring the coordinated cell systems seen in wound healing.	Learners explained, using the Claim-Evidence-Reasoning frame, why each plant cell's shape suits its function; constructed a 'Plant Side' addition to their cumulative specialisation model with directional arrows showing how root hair cells, xylem, phloem, palisade, and guard cells work together; and drew a 'bridge arrow' articulating how plant cell specialisation (particularly photosynthesis in palisade cells) provides the glucose energy that ultimately powers animal cell processes such as fibroblast collagen synthesis during wound healing.
Lesson 10	Annotated microscope images and	Cells of the same type group together to	No single specialised cell — not the

Cell Organisation: Cells → Tissues → Organs → Organ Systems → Organism	diagrams showing one epidermal cell → stratified squamous epithelial tissue → skin organ cross-section → integumentary system → human body; parallel Kenyan school analogy diagram (student → class → Mwangaza Secondary School → Nairobi County Education → Kenyan society).	form tissues; different tissues combine to form organs; organs working together form organ systems; organ systems together constitute the whole organism. In the wound-healing phenomenon, epidermal cells form epithelial tissue, which is part of the skin organ, which belongs to the integumentary system, which protects the entire organism.	platelet, the neutrophil, the fibroblast, nor the epidermal cell — can heal the cut finger alone. Healing requires all four cell types to be organised into tissues, organs, and systems that coordinate their functions. The organisational hierarchy is what transforms individual specialised cells into a living, self-repairing organism — answering the driving question by showing that shape determines job at the cell level, but ORGANISATION determines survival at the organism level.
Lesson 11 Synthesis and Model Revision: Putting the Cell Story Together	In this lesson I observed that learners could integrate their cumulative cell knowledge into a four-zone poster model, that peer feedback during the gallery walk was most precise when learners referenced specific organelles and functions rather than general impressions, and that the DQB closure generated genuine discussion about which questions were fully answered versus partially answered.	Learners consolidated that plant cells and animal cells share core organelles but differ in the presence of a cell wall, large central vacuole, and chloroplasts; that specialised cells in the wound-healing response have structural features directly suited to their functions; and that the organisational hierarchy from cell to organism explains how individual cell functions combine to produce complex responses such as healing.	Learners explained the driving question — why cells look so different and how shape determines function — by constructing an integrated model that connected organelle structure to cell function to tissue and organ roles to whole-organism response, using the Nairobi wound-healing phenomenon as the unifying explanatory context.
Lesson 12 Consolidation, DQB Closure, and Final Explanation — The Healing Cut Fully Explained	The class co-constructed a complete, sequential, cell-by-cell explanation of wound healing on the board — from the moment of the cut to the closure of the skin — naming platelets, neutrophils, macrophages, fibroblasts, and epidermal cells with their shapes and key organelles. A minimum of 8 students contributed publicly via cold-call.	All specialised cells in the body originate from the same organism and carry the same DNA, but different genes are expressed in different cells, producing different structures — and those structures directly determine function. The shape of a cell is not accidental: it is the physical expression of the job that cell is specialised to do. This is the complete answer to the driving question.	The anchoring phenomenon — The Healing Cut — is fully explained: platelets (small, anucleate, sticky) aggregate and initiate clotting; neutrophils (multi-lobed nucleus, abundant lysosomes) engulf bacteria; macrophages (irregular, flexible membrane) engulf debris and dead cells; fibroblasts (elongated, extensive rough ER and Golgi) secrete collagen scaffolding; epidermal cells (tightly packed, active nucleus) divide and migrate to close the surface. Every cell's shape is a solution to a specific biological problem — and that is why cells in the same organism look so completely different from each other.

ATTRIBUTION AND LICENSING

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